Practice Set 5 Paper 3 – Subspecialities of Urology

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Practice paper compiled from the question bank.

Q1. Describe early and late management of PFUD injuries

Type: LAQ (25 marks)

PFUD is taken as pelvic fracture urethral distraction defect, i.e. PFUI (pelvic fracture urethral injury).

Introduction and mechanism

- PFUI occurs in about 10% of male and up to 6% of female pelvic fractures. Highest risk: straddle fracture of all four pubic rami, unstable fractures, symphyseal diastasis.
- The posterior urethra is fixed to the pubis (puboprostatic ligaments, perineal membrane), so displacement shears it, usually at the **bulbomembranous junction**. In prepubertal boys the tear may extend into the prostate and bladder neck.
- The sphincter is avulsed but usually functional, so most men stay continent after repair.
- The gap fills with scar: an obliterative **distraction defect**, not a true stricture.

Classification

System	Types
Campbell-Walsh	**Partial**; **minor distraction** (a cystoscope passes retrogradely into the bladder); **major distraction** (it cannot). Need for urethroplasty rises from partial to major
Goldman	I stretched, intact; II tear above the perineal membrane; III tear of membranous urethra extending below the membrane into the bulb (commonest); IV bladder neck tear into the urethra; IVa bladder base injury mimicking IV; V pure anterior injury

Diagnosis

- Triad: **blood at the meatus, inability to void, palpable bladder**. A "high-riding" prostate is unreliable.
- **Blood at the meatus mandates RUG (retrograde urethrogram)** before catheterisation (oblique, 20 mL contrast). Extravasation with bladder filling means partial injury; without filling, complete.
- Females: vaginal examination and urethroscopy. Cystogram or CT (computed tomography) for bladder injury.

Early management

1. ATLS (Advanced Trauma Life Support) resuscitation; associated injuries take priority.
2. **SPC (suprapubic cystostomy): the standard of care.** Percutaneous (peel-away sheath, 16 Fr Foley; avoid small "punch" tubes) or open through a lower midline incision, which allows bladder inspection.
    - SPC is safe with pelvic plating (AUA [American Urological Association] Urotrauma guideline); tunnel it high, away from hardware.
3. **Primary realignment (PR):** a brief attempt in a stable patient, within days, by antegrade plus retrograde flexible cystoscopy over a guidewire. Catheter out at 4–6 weeks; keep the SPC until voiding is good.
4. Partial tear: one gentle catheter attempt allowed.
5. **Immediate open exploration** only for bladder neck or rectal injury, or laparotomy for other reasons.

- **Also know:** female urethral injuries are usually repaired early.

SPC versus primary realignment

	SPC + delayed urethroplasty	Primary realignment
Advantage	Simple; haematoma untouched	Partial tears may heal; aligned ends ease repair
Drawback	Complete injuries need urethroplasty	Most still form a 1–2 cm stenosis
Evidence	Prospective data: **no significant difference** in need for urethroplasty	Modern PR does not add ED (erectile dysfunction) or incontinence

Late management

Timing: after **3 months** (scar stable), once the patient is ambulant.

Evaluation

- **"Up-and-down-o-gram":** cystogram plus RUG while the patient tries to void, to measure the gap; antegrade flexible cystoscopy via the SPC tract if the bladder neck stays closed. Bladder neck appearance does **not** predict its later function.
- **MRI (magnetic resonance imaging):** gap length, prostatic displacement (cephalad displacement predicts pubectomy).

Endoscopic: DVIU (direct vision internal urethrotomy) only for short, incomplete stenosis. "Cut-to-the-light" for gaps ≥1 cm fails (false passages).

Perineal anastomotic urethroplasty (treatment of choice)

1. Exaggerated lithotomy (under 5 hours); sounds via meatus and SPC tract.
2. Midline or lambda incision; mobilise the bulbar urethra up to the penoscrotal junction (not beyond: chordee).
3. Divide the urethra at the scar; excise **all** fibrosis until healthy prostatic mucosa appears over the sound.
4. Spatulate both ends to 30 Fr; 8–10 interrupted 4-0 absorbable sutures; tension-free mucosa-to-mucosa anastomosis over a silicone catheter.

Progressive perineal approach (each step shortens the path to bridge a longer gap):

Step	Manoeuvre
1	Bulbar urethral mobilisation (enough for most)
2	Separation of the corporal bodies (intracrural space)
3	Inferior wedge pubectomy (Kerrison rongeur)
4	Supracrural rerouting of the urethra around one corpus

- **Abdominoperineal or transpubic approach:** urethrorectal fistula, failed repair, bladder neck injury, children.
- **Postoperative:** voiding study at 21–28 days; cystoscopy at about 6 months.

Outcomes and complications

- Success above 95%. Restenosis 5–15%; often DVIU-treatable because the scar is already excised.
- **ED** up to 82%, mostly from the injury (cavernous nerves, pudendal arteries). **Incontinence <4%** (bladder neck). Follow ≥1 year.
- **Also know:** chordee, penile shortening, ejaculatory dysfunction, lithotomy neuropathy, penile Doppler before repair in impotent men.

Diagram to draw: sagittal pelvis showing the bulbomembranous gap and the four progressive perineal steps.

Key points to remember:

- Blood at meatus → RUG; SPC is the standard initial step.
- Primary realignment: brief, endoscopic; no difference in final urethroplasty rates.
- Delayed perineal anastomotic urethroplasty after 3 months; bulbar mobilisation → crural separation → pubectomy → rerouting.
- ED mostly from the injury; incontinence <4%.

Q2. Describe in detail TCC of ureter.

Type: LAQ (25 marks) (also answered for the whole upper tract in 2023-Winter Paper 2, Q1 and 2024-Winter Paper 3, Q1)

TCC (transitional cell carcinoma) of the ureter is taken as ureteric UTUC (upper tract urothelial carcinoma).

Introduction and epidemiology

- UTUC is 5–10% of urothelial cancers; pelvicalyceal tumours are about twice as common as ureteric ones. Incidence is rising, **driven by ureteric tumours**.
- Peak age 70–90; male:female 2:1. About 60% are invasive at diagnosis, because the upper tract wall is thin.
- **Site: distal ureter 70%, mid 25%, proximal 5%**, explained by downstream implantation.

Aetiology and pathogenesis

Factor	Key point
Tobacco	Commonest; risk 2.5–7-fold (aromatic amines)
Aristolochic acid	Balkan and Chinese herb nephropathy; A:T → T:A transversions in TP53; often bilateral
Lynch syndrome	Mismatch repair genes (MSH2 highest risk); screen if young or family history
Occupational	Aniline dyes, benzidine, β-naphthylamine
Phenacetin	Papillary necrosis (history in 11% of ureteric tumours)

- **Also know:** arsenic, cyclophosphamide, chronic stones (squamous carcinoma).
- **Intraluminal seeding** explains distal predominance and bladder recurrence (22–47%); **field change** explains multifocality.

Pathology and spread

- 85% papillary, 15% sessile; WHO (World Health Organization) 2004/2016 low or high grade. Ureteric tumours are low grade and low stage in 55–75%.
- **Why ureteric tumours do worse:** the renal parenchyma is a barrier for pelvic tumours, but periureteric fat holds a rich lymphovascular plexus. At equal stage, ureteric tumours have more local and distant failure.
- Spread: direct, lymphatic (**mid and distal ureter → pelvic nodes**; upper ureter → hilar and retroperitoneal), blood (lung, liver, bone).

Clinical features

- **Haematuria** (56–98%); **flank pain** (20–32%), dull from obstruction or colic from clots; incidental in about 15%.

Investigations

- **CTU (computed tomography urography):** best test (sensitivity ~92%, specificity ~95%). Filling defect, hydronephrosis, ureteric wall thickening. Tumour ~46 HU (Hounsfield units) versus uric acid stone ~100 HU.
- **Hydronephrosis predicts invasive ureteric tumour in up to 80%.**
- **Cystoscopy** always (concurrent bladder cancer 17%). **Selective ureteric cytology**, taken before contrast.

- **Retrograde pyelography: goblet (champagne-glass) sign**, the ureter dilated just below the tumour; **Bergman's sign**, the catheter coiling below it.
- **URS (ureteroscopy) with biopsy:** grade correct in ~90% but **stage cannot be assessed**. URS before nephroureterectomy raises bladder recurrence.
- **Also know:** MRU (magnetic resonance urography) if contrast is contraindicated; renogram.

Staging (TNM [tumour, node, metastasis] 8th edition)

T	Ureter
Ta / Tis	Papillary non-invasive / carcinoma in situ
T1	Subepithelial connective tissue
T2	Muscularis
T3	Periureteric fat
T4	Adjacent organs

- N1 single node ≤2 cm; N2 >2 cm or multiple; M1 distant.

Risk stratification (EAU, European Association of Urology)

Low risk: all needed	High risk: any one
Unifocal, <2 cm	Multifocal or ≥2 cm
Low-grade cytology and biopsy	High-grade cytology or biopsy
No invasion on CTU	Hydronephrosis; invasion on imaging
—	Previous cystectomy; variant histology

Management

A. Kidney-sparing: elective in low risk; imperative in solitary kidney, bilateral disease, CKD (chronic kidney disease) or unfit patients.

- **Ureteroscopic laser ablation** (holmium or thulium) after biopsy; avoid circumferential ablation (stricture).
- **Segmental ureterectomy:** first examine the whole tract above for multifocality; pelvic node dissection if high grade. Gives a staging specimen. Lower-ureteric tumours with unfavourable features need resection, not endoscopy.

Site	Operation
Distal ureter	Distal ureterectomy with bladder cuff and reimplantation; psoas hitch (~5 cm extra) or Boari flap (10–15 cm)
Mid or proximal	Excision with 1–2 cm margins (frozen section) and ureteroureterostomy (rare)
Whole ureter	Ileal ureter or autotransplantation

- Refluxing reimplant eases surveillance; non-refluxing limits seeding (debated).
- Adjuvant upper-tract BCG (bacillus Calmette–Guérin) or MMC (mitomycin C).

B. RNU (radical nephroureterectomy) with bladder cuff: standard for high risk.

- Kidney outside Gerota's fascia; clip the ureter early; **remove the intramural ureter and cuff** (stump recurrence 30–75% if left). Open, laparoscopic and robotic routes are equivalent; "pluck" and stapling techniques are abandoned.
- Lymphadenectomy: pelvic nodes for lower ureteric tumours.
- **Single postoperative intravesical MMC** (ODMIT-C): bladder recurrence 28% → 16% at 1 year.

C. Perioperative chemotherapy: POUT (2020), adjuvant platinum for pT2–T4 or N+, improved disease-free survival. Neoadjuvant chemotherapy when post-RNU renal function would exclude cisplatin.

D. Metastatic: enfortumab vedotin + pembrolizumab (EV-302); gemcitabine–platinum with avelumab maintenance.

Prognosis and follow-up

- 5-year disease-specific survival: localised ~89%, regional ~63%, distant ~17%. Stage is the strongest factor.
- Cystoscopy from 3 months and yearly CTU after RNU; closer URS and CTU after kidney-sparing treatment.

Diagram to draw: distal ureterectomy with psoas-hitch reimplantation, beside the goblet sign.

> **Key points to remember:**
>
> - Ureteric tumours: 70% distal; hydronephrosis predicts invasion (up to 80%).
> - Ureteric T3 does worse than pelvic T3 (no parenchymal barrier).
> - URS biopsy gives grade, not stage; distal ureterectomy for selected distal tumours.
> - RNU + cuff + single MMC (ODMIT-C); adjuvant platinum (POUT).

Q3a. Early vs. delayed repair in VVF

Type: SAQ (10 marks) (also answered in 2021-Summer Paper 2, Q3b)

Introduction

- VVF (vesicovaginal fistula): an epithelialised bladder–vagina tract causing continuous leakage.
- Causes: **obstetric** (obstructed labour) in low-resource settings; **iatrogenic** (mainly hysterectomy) in high-resource settings; radiation; malignancy. Common mechanism: **ischaemic necrosis**.

The debate

	Early repair	Delayed repair
Definition (varies)	Days, <6 weeks or <3 months	Traditionally ≥3 months
Rationale	Clean iatrogenic tissue heals well; ends wetness and distress early	Inflammation settles, necrosis sloughs, margins become clear
Evidence	**Comparable closure** in most series; Waaldijk: immediate obstetric repair gave very high first-attempt closure	One small series: more recurrence when repaired <6 weeks
Drawback	Friable tissue if inflamed	Prolonged distress, no better outcome

- Some authors avoid the **4–12 week** window (active healing).
- **EAU (European Association of Urology) consensus:** no minimum delay if there is **no active infection and no recent irradiation**.

Timing by cause

- **Clean post-hysterectomy:** early repair.
- **Obstetric with necrotic edges:** catheter trial, then repair at about 3 months.
- **Radiation:** wait **≥12 months** after radiotherapy; biopsy to exclude recurrence; interposition flap; consider diversion if the bladder is poor.

Conservative trial while waiting

- Catheter drainage cures about 15%. Best if small (<3 mm best; <1 cm), fresh, non-irradiated and catheterised **within 3 weeks**. Once epithelialised (≥6 weeks) closure is unlikely. EAU: if tried, at least 12 weeks.
- **Also know:** fulguration of tiny tracts; fibrin glue (not recommended by EAU).

Principles at repair

- Treat infection; ureteric stents if within 1 cm of a ureter.
- Vaginal or abdominal route (open, laparoscopic, robotic); tension-free, multilayer, watertight closure; **interposition flap** (Martius, omentum, peritoneum) if irradiated or recurrent; catheter drainage for about 2 weeks.

Diagram to draw: timeline showing early window, 4–12 week zone, 3 months (obstetric) and 12 months (radiation).

> **Key points to remember:**
>
> - Early repair for clean iatrogenic VVF; closure comparable to delayed.
> - Delay ~3 months for obstetric, ≥12 months for radiation fistula.
> - Catheter trial works only if small, fresh and started within 3 weeks.

Q3b. Maldescended testes

Type: SAQ (10 marks) (algorithm also in 2018-Summer Paper 3, Q3a; bilateral cases in 2024-Winter Paper 1, Q2)

Taken as any testis not in a stable scrotal position: undescended, ectopic, retractile and acquired.

Definitions

Type	Description
Undescended (cryptorchid)	Arrested on the normal path: abdominal, canalicular, prescrotal
Ectopic	Off the normal path: **superficial inguinal pouch** (commonest), perineal, femoral, prepubic, penile, transverse (crossed)
Retractile	Scrotal testis pulled up by the cremaster; stays down after traction
Acquired (ascending)	Previously scrotal; presents at 8–11 years
Vanished / agenesis	Lost in utero (torsion) / never formed

Epidemiology and aetiology

- 1–4% of term boys, more in preterm; most descend by 6 months. 75–80% palpable; 60–70% unilateral.
- Descent needs **INSL3 (insulin-like peptide 3)** and **androgen** acting on the gubernaculum. Risk factors: prematurity, low birth weight, syndromes (prune belly, cerebral palsy).

Evaluation

- Warm room, supine and cross-legged; sweep from the anterior superior iliac spine to the scrotum to find the lowest position. Sustained traction fatigues the cremaster and separates retractile from undescended.
- **No ultrasound or MRI (magnetic resonance imaging)**: ultrasound sensitivity 45% for non-palpable testes.
- **Bilateral non-palpable or with hypospadias:** karyotype, 17-OHP (17-hydroxyprogesterone), AMH (anti-Müllerian hormone) to exclude **CAH (congenital adrenal hyperplasia)** and other DSD (differences of sex development).

Management

- Observe to 6 months (corrected age); **orchidopexy from 6 months, preferably by 12, by 18 months at the latest**. No hormones (LHRH [luteinising hormone-releasing hormone] works in about 20%).

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Palpable (undescended or ectopic) → inguinal or scrotal (Bianchi) orchidopexy
Non-palpable → examination under anaesthesia → laparoscopy
   testis seen        → laparoscopic orchidopexy / Fowler–Stephens
   vessels enter ring → inguinal exploration, excise nubbin
   blind-ending vessels → stop (vanished)
Retractile → yearly review
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- Postpubertal intra-abdominal testis, normal contralateral: orchidectomy.

Sequelae

- **Infertility:** paternity about four-fifths after unilateral, one-third after bilateral.

- **Cancer:** 2–5-fold TGCT (testicular germ cell tumour) risk, higher with late surgery; teach self-examination.
- **Also know:** hernia, torsion, trauma.

Diagram to draw: line of descent with ectopic sites.

> **Key points to remember:**
>
> - Ectopic = off the path; commonest site superficial inguinal pouch.
> - Orchidopexy 6–12 months, by 18 months.
> - No imaging, no hormones; laparoscopy for non-palpable.
> - Bilateral non-palpable = DSD work-up first.

Q3c. Extended indications of Shodgrass repair in hypospadias

Type: SAQ (10 marks)

"Shodgrass" is taken as Snodgrass TIP (tubularised incised plate) repair. Technique: see Practice Set 5 Paper 3, Q3d.

Introduction

- Snodgrass (1994): a deep midline incision widens the urethral plate, which is tubularised; the incision re-epithelialises. Designed for distal hypospadias, it now covers **over 90% of distal and about 80% of midshaft repairs**.

Prerequisites for any extended use

- Curvature **<30° after degloving**, corrected by dorsal plication **without dividing the plate**.
- Supple, well-vascularised plate of adequate width and depth. **POST (plate objective scoring tool):** a low AB/BC ratio means a poor plate, favouring an inlay graft.
- **Glans width ≥14 mm**, or preoperative testosterone to enlarge it.

Extended indications

Indication	Rationale	Caveat
Midshaft hypospadias	Same principle	Good results
Proximal hypospadias with mild curvature, good plate	Single stage; no flaps	More fistula, stricture, recurrent curvature; Campbell-Walsh authors now prefer staging
Redo or failed repairs with preserved plate	Plate often survives	Worse than primary; inlay graft if scarred
MIP (megameatus intact prepuce)	Wide, deep plate	GAP (glans approximation procedure) alternative
Narrow or shallow plate: TIP + **dorsal inlay graft** ("Snodgraft")	Preputial or buccal graft fills the incision, so it cannot contract	Less meatal stenosis
Adults, circumcised boys	No preputial tissue needed	Success similar

- **Also know:** in redo cases with poor blood supply a dorsal buccal inlay beats onlay or tube; incise-and-graft is also used in adult urethral stricture (Asopa dorsal inlay) and female urethroplasty.

Contraindications

- Curvature >30° needing plate division; absent or scarred plate; penoscrotal transposition. Use staged repair (Bracka, STAG [staged tubularised autograft], Byars).

Outcomes

- Pfistermuller meta-analysis (2015): complications 10.6%, fistula 5.7%, meatal stenosis 3.6%; **proximal and redo cases did significantly worse**. A dartos barrier layer improves results.

Diagram to draw: cross-sections: plate incision, inlay graft, tubularisation.

Key points to remember:

- Extended to midshaft, selected proximal, redo, MIP and adults.
- Key condition: curvature <30° without plate division.
- Poor plate → dorsal inlay graft ("Snodgraft").
- Proximal and redo TIP fail more often.

Q3d. Snodgrass Repair for distal hypos.

Type: SAQ (10 marks) (also answered in 2023-Winter Paper 3, Q3c)

"Distal hypos." is taken as distal hypospadias.

Introduction

- **TIP (tubularised incised plate) urethroplasty**: Snodgrass 1994, after Rich's hinging incision (1989). Used for over 90% of distal repairs.
- Distal hypospadias forms 70–85% of cases.

Principle

- A **deep midline relaxing incision** of the urethral plate widens a narrow plate so that it can be tubularised without tension.
- The raw area **heals by re-epithelialisation** without a graft, giving a **vertical slit-like meatus**.

Indications and assessment

- Distal hypospadias with curvature <30° after degloving, and an adequate plate and glans.
- **Glans width <14 mm** raises complications: give preoperative testosterone.
- **POST (plate objective scoring tool):** a low ratio favours an inlay graft.
- Timing: 6–12 months of age.

Operative steps

1. Glans stay suture; tourniquet.
2. U-incision around the plate and meatus; circumscribe and deglove; artificial erection.
3. Separate the glans wings deeply from the plate.
4. **Midline incision of the plate** from inside the meatus to the end of the plate, down to the corpora.
5. Tubularise over a **6–8 Fr stent** with subepithelial 6-0/7-0 absorbable sutures, distal to proximal; the first stitch about 3 mm from the plate end leaves an oval meatus.
6. **Dartos flap** (dorsal preputial) over the neourethra as a barrier.
7. Two-layer glansplasty; mucosal collar; skin closure (Byars flaps) or preputioplasty.
8. Stent 5–7 days; compressive dressing; oxybutynin.

Complications and outcomes

- Pfistermuller meta-analysis (2015, 4,675 boys): complications 10.6%, reoperation 4.5%; **fistula 5.7%**, meatal stenosis 3.6%, stricture 1.3%.
- **Meatal stenosis is commoner than after Mathieu**; glans dehiscence also occurs.
- Distal repair success 83–95%.

Modifications

- **Dorsal inlay graft** (inner preputial skin) for a shallow plate; lateral plate incisions to 12 mm width.
- **Also know:** MAGPI (meatal advancement and glanuloplasty incorporated), Mathieu, Thiersch–Duplay, onlay flap.

Diagram to draw: cross-sections showing plate incision, tubularisation and dartos cover.

Key points to remember:

- Snodgrass 1994; incise, tubularise, heals by epithelialisation.
- Slit-like meatus; dartos barrier reduces fistula.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).

Q3e. Mixed gonadal dysgenesis.

Type: SAQ (10 marks) (DSD management in general in 2019-Summer Paper 3, Q3f)

Definition

- MGD (mixed gonadal dysgenesis): a **dysgenetic testis on one side and a streak gonad on the other**, with persistent Müllerian structures and variable virilisation.
- A sex chromosome DSD (difference of sex development), usually **45,X/46,XY mosaicism** (about 1 in 5,000 births). **Second commonest cause of neonatal atypical genitalia after CAH (congenital adrenal hyperplasia).**

Pathogenesis

- The Y-bearing line forms a testis; the 45,X line leaves a streak.
- AMH (anti-Müllerian hormone) and testosterone act **locally**, so anatomy is asymmetric: Wolffian duct on the **testis side**; tube and hemiuterus on the **streak side**.
- 45,X/46,XY spectrum: female (Turner + Y), atypical (MGD) or male with normal testes (about 85%).

Clinical features

- **Asymmetric genitalia:** phallus with chordee, perineal hypospadias, urogenital sinus; one testis in a hemiscrotum, the other side impalpable. Hernia may contain a uterus or tube.
- **Turner features:** short stature, coarctation, horseshoe kidney.

Evaluation (MDT, multidisciplinary team)

- **Karyotype** (many cells, for mosaicism); **17-OHP (17-hydroxyprogesterone)** to exclude CAH; LH (luteinising hormone), FSH (follicle-stimulating hormone), testosterone, AMH.
- **Pelvic ultrasound** (uterus, asymmetric gonads); genitography, endoscopy; **laparoscopy with gonadal biopsy**. No gonadectomy during diagnosis.
- Echocardiography, renal ultrasound (Turner screening).

	MGD	Ovotesticular DSD
Karyotype	45,X/46,XY	Mostly 46,XX
Gonads	Dysgenetic testis + streak	Ovary + testis, or ovotestis
Tumour risk	15–35%	About 3%

Management

- **Sex of rearing:** individualised; more androgen exposure predicts male identity; dysphoria is commoner when raised female. Avoid irreversible infant surgery.
- **Gonads:** germ cell neoplasia 15–35% (gonadoblastoma, then dysgerminoma or seminoma) in streak and testis alike. **Remove streaks** (biopsy first if tissue uncertain).
  - Raised male: keep a scrotal testis under surveillance; bring down an intra-abdominal one if possible.
  - Raised female: remove or relocate the testis.
- **Surgery:** male: staged hypospadias repair, orchidopexy; female: feminising surgery deferred.
- **Hormones:** pubertal sex steroids; growth hormone. Infertility usual.

Diagram to draw: asymmetric anatomy: testis with vas; streak with tube and hemiuterus.

Key points to remember:

- 45,X/46,XY; testis + streak + Müllerian remnants; asymmetric.
- Second commonest cause of atypical genitalia after CAH.
- Tumour risk 15–35%: remove streaks.
- Screen for Turner features.

Q3f. Hemostasis in laparoscopic surgery.

Type: SAQ (10 marks) (also answered in Practice Set 2 Paper 4, Q3f; partial nephrectomy haemostasis in 2019-Summer Paper 3, Q1)

Principles

- Devices seal by **denaturing protein**. Plan from imaging; keep a **haemorrhage tray** (laparoscopic Satinsky, 10-mm suction, vascular suture with Lapra-Ty, haemostatic agent).

Effect of pneumoperitoneum

- CO_2 at 12–15 mmHg **tamponades venous bleeding**. Before exit, **lower pressure to about 5 mmHg** and inspect the operative site and each port.

Energy devices

Device	Lateral thermal spread	Note
Monopolar	5–7 mm	Capacitive coupling, insulation failure, direct coupling
Bipolar	1–6 mm	Current only between the jaws
Vessel sealer (LigaSure)	1–3 mm	Seals vessels **≤7 mm**; one application only
Ultrasonic (Harmonic)	≈1 mm	No current; blade >200 °C, cools slowly
Argon beam	Superficial	Raw surfaces; gas embolism risk

Clips

- **Titanium clips:** no diathermy near them (slip).
- **Hem-o-lok** (locking polymer): circumferential dissection, see the tip beyond the vessel, feel the snap, at least two clips on the patient side with a 2-mm cuff. The manufacturer **contraindicates it on the living-donor renal artery** after stump haemorrhages.
- **Lapra-Ty** absorbable clips act as knots (renorrhaphy).

Staplers

- Endo GIA via a 12-mm port; **grey (2.0 mm) or white (2.5 mm)** loads for renal vessels; blue or green for thick tissue.
- **Never fire across clips.**

Sutures and haemostatic agents

- Intracorporeal or barbed sutures; fibrin sealant, gelatin–thrombin matrix, oxidised cellulose.

Managing bleeding

- **Venous injury:** raise pressure to 25 mmHg → suction to identify → pledget tamponade → sealant → extra 5-mm port for suturing or Satinsky → hand-assist → convert. Keep the patient well filled (gas embolism).
- **Arterial:** minor, tamponade; renal artery, staple; aorta, convert.
- **Port-site bleeding:** cantilever the trocar to find it; diathermy or transfascial suture. Blunt trocars reduce it.

Diagram to draw: Hem-o-lok placement with two patient-side clips and a 2-mm cuff.

Key points to remember:

- Pneumoperitoneum hides venous bleeding: inspect at low pressure before exit.
- LigaSure seals ≤7 mm; spread monopolar 5–7 mm, ultrasonic ≈1 mm.
- Hem-o-lok contraindicated on the donor renal artery.
- Never staple across clips; grey or white loads for vessels.

Rapid Revision Sheet – Practice Set 5 Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe early and late management of PFUD injuries (LAQ)

- Blood at meatus → RUG; SPC is the standard initial step.
- Primary realignment: brief, endoscopic; no difference in final urethroplasty rates.
- Delayed perineal anastomotic urethroplasty after 3 months; bulbar mobilisation → crural separation → pubectomy → rerouting.
- ED mostly from the injury; incontinence <4%.

Q2. Describe in detail TCC of ureter. (LAQ)

- Ureteric tumours: 70% distal; hydronephrosis predicts invasion (up to 80%).
- Ureteric T3 does worse than pelvic T3 (no parenchymal barrier).
- URS biopsy gives grade, not stage; distal ureterectomy for selected distal tumours.
- RNU + cuff + single MMC (ODMIT-C); adjuvant platinum (POUT).

Q3a. Early vs. delayed repair in VVF (SAQ)

- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months for obstetric, ≥12 months for radiation fistula.
- Catheter trial works only if small, fresh and started within 3 weeks.

Q3b. Maldescended testes (SAQ)

- Ectopic = off the path; commonest site superficial inguinal pouch.
- Orchidopexy 6–12 months, by 18 months.
- No imaging, no hormones; laparoscopy for non-palpable.
- Bilateral non-palpable = DSD work-up first.

Q3c. Extended indications of Shodgrass repair in hypospadias (SAQ)

- Extended to midshaft, selected proximal, redo, MIP and adults.
- Key condition: curvature <30° without plate division.
- Poor plate → dorsal inlay graft ("Snodgraft").
- Proximal and redo TIP fail more often.

Q3d. Snodgrass Repair for distal hypos. (SAQ)

- Snodgrass 1994; incise, tubularise, heals by epithelialisation.
- Slit-like meatus; dartos barrier reduces fistula.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).

Q3e. Mixed gonadal dysgenesis. (SAQ)

- 45,X/46,XY; testis + streak + Müllerian remnants; asymmetric.
- Second commonest cause of atypical genitalia after CAH.
- Tumour risk 15–35%: remove streaks.
- Screen for Turner features.

Q3f. Hemostasis in laparoscopic surgery. (SAQ)

- Pneumoperitoneum hides venous bleeding: inspect at low pressure before exit.
- LigaSure seals ≤7 mm; spread monopolar 5–7 mm, ultrasonic ≈1 mm.
- Hem-o-lok contraindicated on the donor renal artery.
- Never staple across clips; grey or white loads for vessels.